SPEEDRUN SR008 / NOVEMBER 2026

STEPLOCK

VISUAL PROCEDURE ASSURANCE

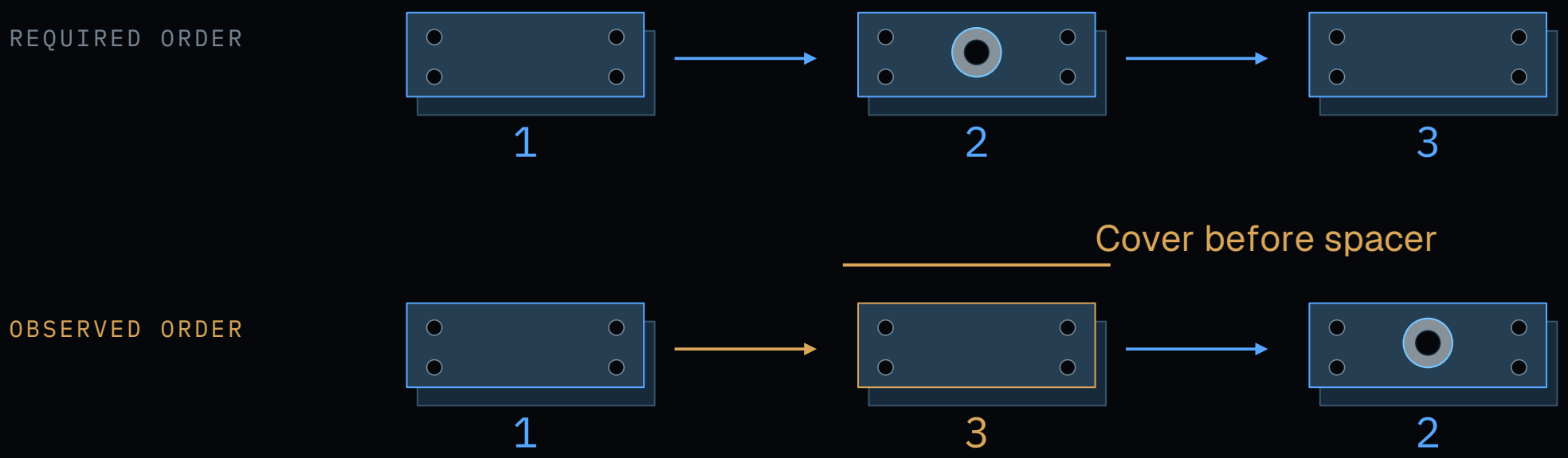
Catch the *wrong move* before the next step

Visual procedure verification for smaller manufacturers

SAISHREEK SINGH / SOLE FOUNDER



The mistake happens now.
We find it later.



The same parts can produce the wrong procedure.

Every later step can add to the cost of fixing it.





The next operation can hide the error.

An earlier placement check could let the operator correct it before closing the assembly.

Quality control should fit the factory using it.

Origin: a friend’s family factory. A few supervisors held the procedural know-how.

RESOURCE-RICH INSTALLATION

Dedicated engineering.
Integrated inspection hardware.
Application-specific setup.

OUR STARTING CUSTOMER / HYPOTHESIS

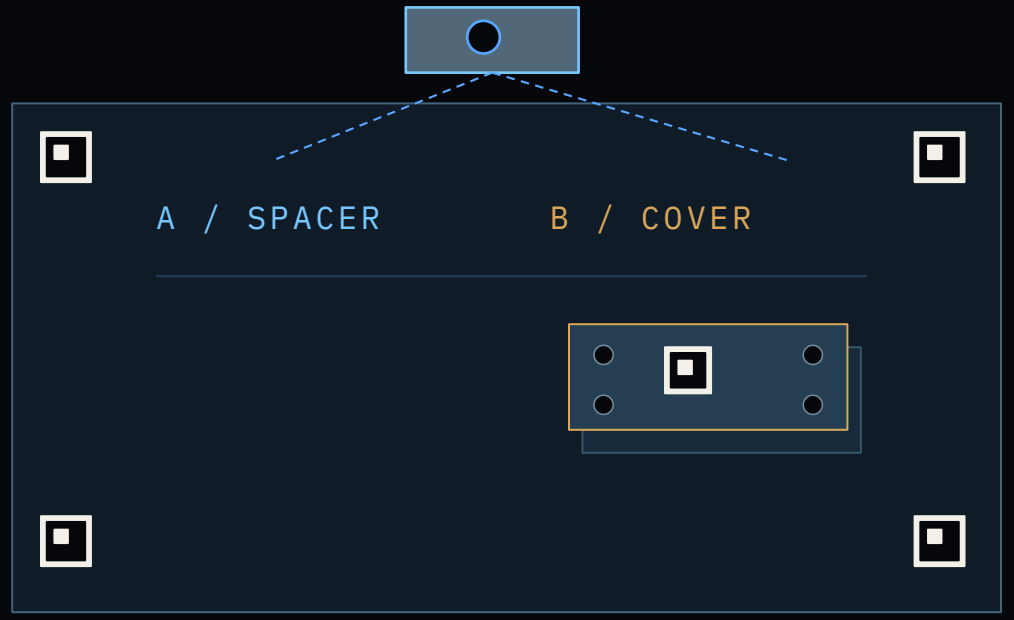
A repeatable assembly bench.
A few supervisors.
A need for manageable setup and cost.



Illustrative stock workshop; not the founder’s factory.
Location unspecified. No affiliation or endorsement.

Sources: Ministry of MSME RAMP; photo Andrea Piacquadio / Pexels, illustrative workshop

What if the process could verify its next step?



HOLD / B REQUIRES A

Cover before spacer

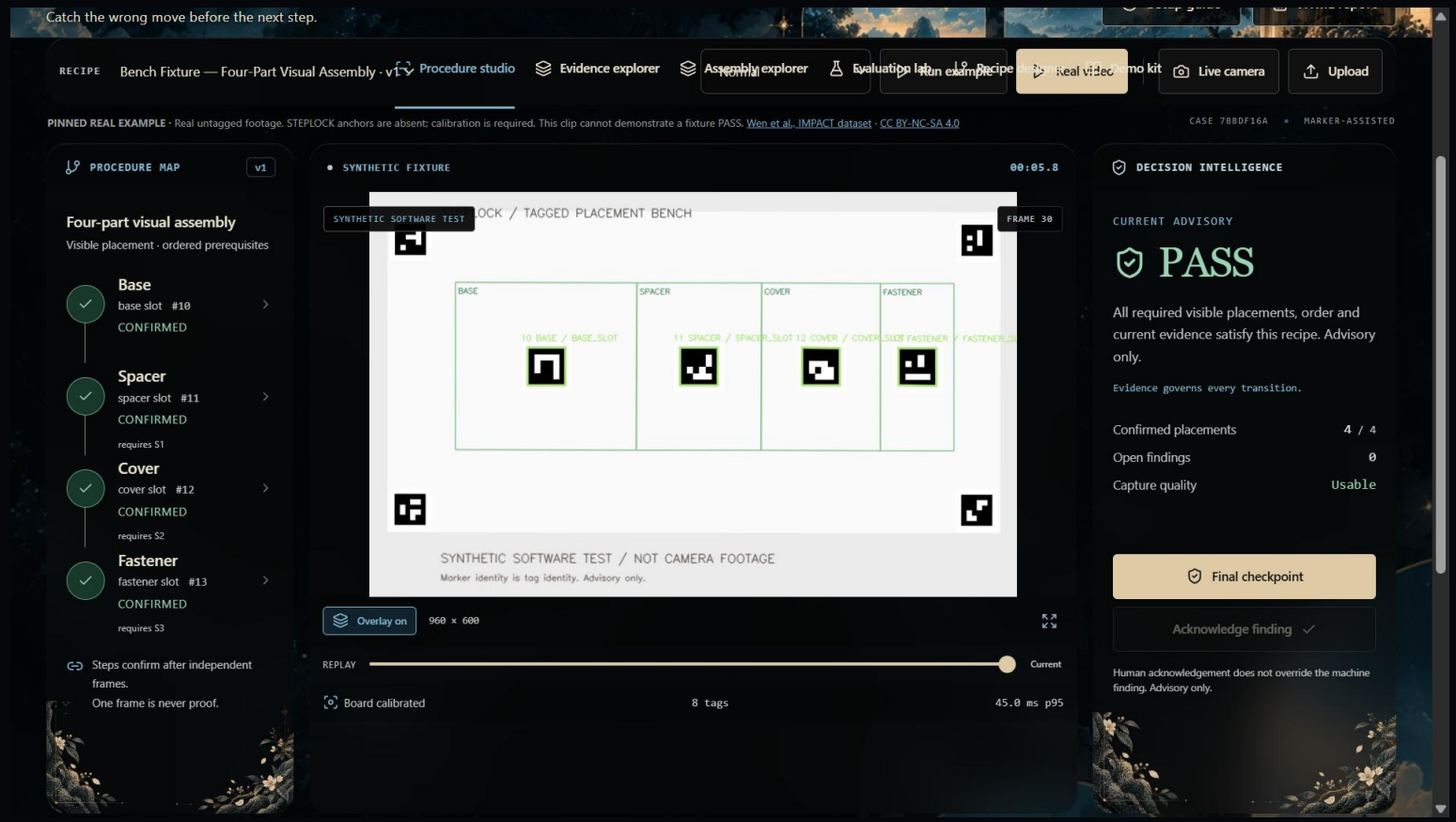
The engine checks earlier supported observations before confirming the next placement.

Observe → Verify → Alert → Correct → Keep evidence

Meet STEPLOCK

Procedure Studio

Visible placements. Procedure checks. Linked evidence.



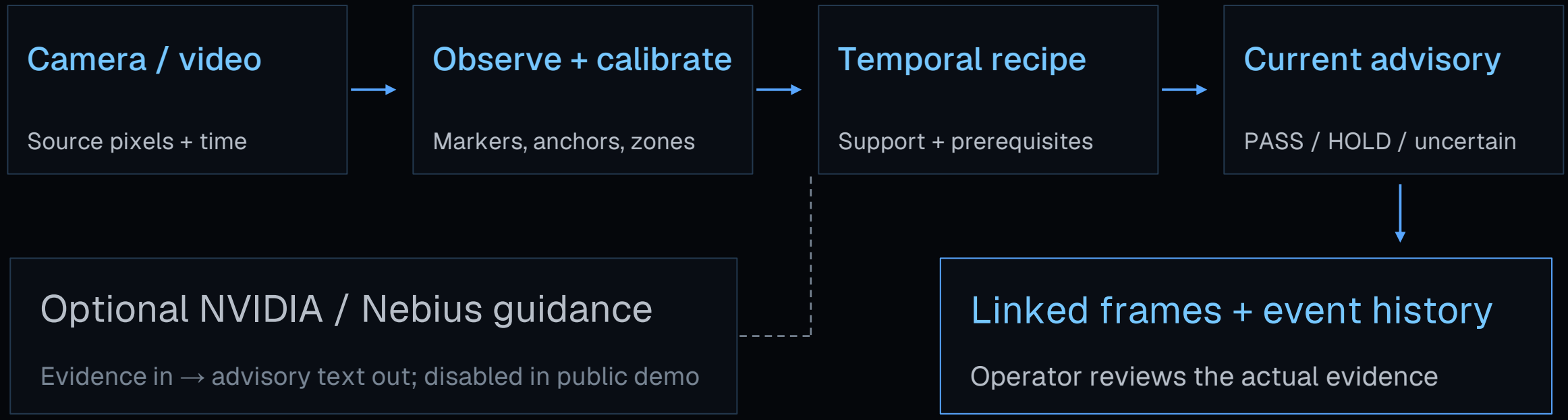
ACTUAL UI

PASS

4 visible placements confirmed

Generated fixture. Not camera footage.

Not just a warning.
An explanation you can inspect.

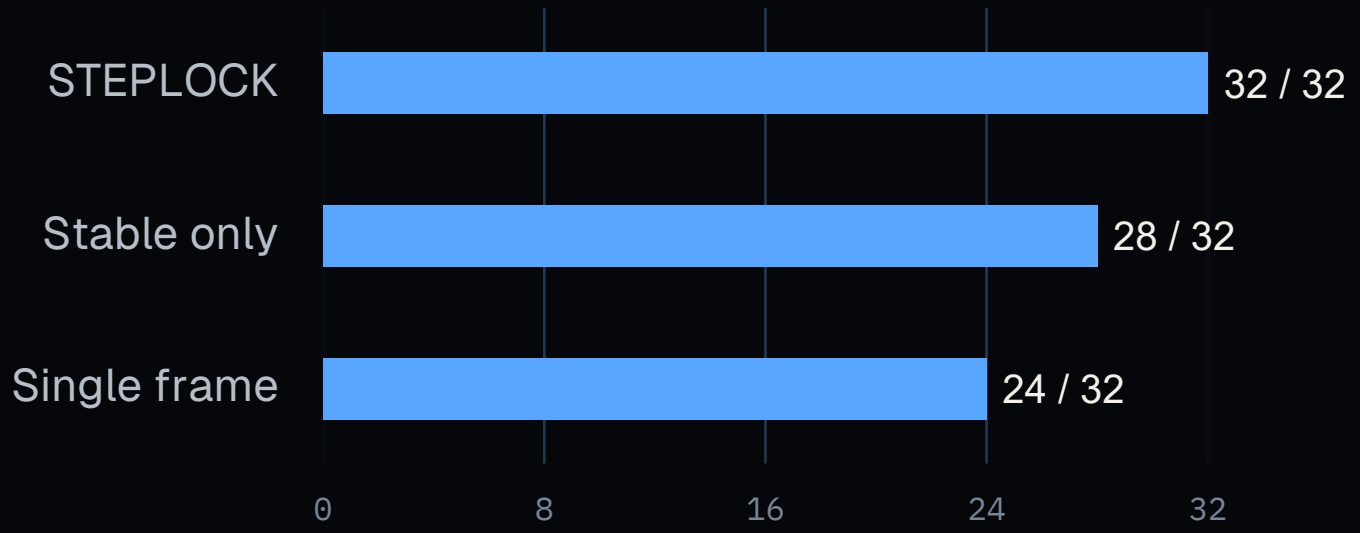


OBSERVED FACTS → EXPLICIT RULES → TRACEABLE DECISION

Source: architecture, procedure engine and NVIDIA/Nebius integration / Optional advice never owns the verdict

Built. Tested. Measured.

CORRECT FINAL STATUS / SAME 32 CLIPS



32/32

correct final statuses

0/24

false PASS

4/4

corrections

CONTROLLED GENERATED FIXTURES. NOT INDUSTRIAL ACCURACY.

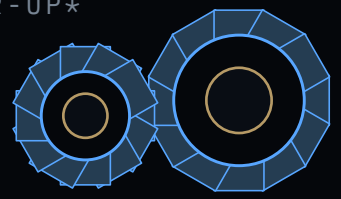
Source: saved 29 Oct 2026 benchmark / 32 generated test clips / 864 frames / Not industrial accuracy

Machines first.
Software next. STEPLOCK now.

Saishreek Singh

SOLE FOUNDER / MECHATRONICS

STEPLOCK / HACKATHON RUNNER-UP*



Robotics taught precision

Robotics club president; World Robot Olympiad India gold.

Software turned it into a product

Independent builds: Peekolitix, Warscythe and SATQuery AI.

≈20 hours/week while studying. Ready to build full-time.

Start where the need is visible.

WHO WE WOULD QUALIFY FIRST

01 Repeatable manual assembly

02 Tags and a fixed camera are practical

03 A supervisor owns the procedure

04 Downstream rework can be measured

A SCALE SCENARIO / NOT TAM

1,000

hypothetical eligible factories

× 2 stations × ₹3,000/month

= ₹7.2 crore ARR

The eligible account count, willingness to pay and installation economics still need validation.

Earn one repeatable deployment. Then earn the next.

01

Supervised pilots

One suitable tagged workflow.
Freeze manual ground truth.

02

Measure the station

Misses, false alerts, setup time, operator acceptance, rework.

03

Package installation

Repeatable recipe, camera setup, training and support.

04

Expand distribution

Manufacturing clusters and implementation partners.

Proposed business model: per-station software subscription + paid setup/support

From detecting mistakes to anticipating them.



TODAY	NEXT	LATER	LONG TERM
Tagged procedure verification	Factory validation + more workflows	Broader observation + process analytics	Predictive process intelligence
Implemented prototype	Planned	Research direction	Long-term ambition

A trusted record of what happened could become a foundation for learning what fails next.

Earlier correction has an economic mechanism to test.

Input / annual outcome	Conservative	Base	Optimistic
Workstations	4	4	4
Errors / station / week	2	4	6
Detected early (assumed)	35%	50%	65%
Rework minutes avoided	15	30	45
Potential hours saved	34 h	192 h	562 h
Gross annual value	₹13k	₹77k	₹225k
First-year net after assumed cost	−₹79k	−₹15k	₹133k

Hours = stations × weekly errors × early detection × minutes avoided × weeks ÷ 60

48 weeks; ₹200/hour; material ₹50/100/150 per early incident; subscription ₹72k/year + ₹20k setup.

STEPLOCK

VISUAL PROCEDURE ASSURANCE

Quality shouldn’t begin *at the end.*

Move manufacturing verification to the moment work happens.

Saishreek Singh · Sole founder

Ready to build full-time and relocate to San Francisco if selected.

Open the working demo ↗

github.com/Yoriichi-GOD/steplock ↗



What the 32/32 result means

Same observations. Three rule sets.

32 test clips; 864 frames.
16 separate development clips.
All arms receive the same frame stream.

Arm	Final status	False PASS
Single frame	24 / 32	8 / 24
Stable only	28 / 32	4 / 24
STEPLOCK	32 / 32	0 / 24

The corpus shares a renderer and a four-part recipe. These results establish controlled software behavior; they do not estimate factory accuracy.

Corrected outcomes: 4/4. Scored events: TP 60, FP 0, FN 0. Physical capture-to-visible latency remains unmeasured.

Real input tested.
Industrial recognition still open.

2,700

real source frames processed

45/45

windows: calibration required

What this proves

Real-video ingestion and safe refusal.
Three executions; correlated windows.

What must come next

Consented tagged trials, independent labels, miss/false-alert rates and setup economics.

Source: IMPACT physical-video compatibility experiment / No dataset frames redistributed

A narrow entry point in an existing market

Approach	Published / typical role	Practical deployment consideration
Manual inspection	Operator/supervisor reviews work	Timing and coverage depend on the workflow
Cognex vision	Visual assembly verification	Application imaging, hardware and configuration
Tulip	Guided inspections, work instructions, records	Station apps and configured integrations
Siemens Inspekto	Ready-to-use AI visual inspection	Dedicated visual-QA system and application fit
STEPLOCK prototype	Tagged temporal rules + linked evidence	Fixed board/tags; physical validation pending

Our hypothesis: make a limited class of procedure checks easier to start and review.

A model to challenge, not a forecast

Market context is not addressability

≈2.36m unincorporated manufacturing establishments with hired workers (ASUSE 2023–24).

Not an addressable-market estimate.

Eligible tagged assembly stations remain uncounted.

Financial model inputs

48 weeks/year; ₹200/hour labor value.
Materials avoided: ₹50 / ₹100 / ₹150.

ROI model: ₹1,500/station/month + ₹20k setup.
Scale sensitivity: ₹3,000/station/month.

Validation gates: willingness to pay · installation time · support burden · measured rework avoided

Sources behind the story

01 / Product + evidence	Public code, architecture, limitations, source PNG provenance
02 / Technical measurements	Saved benchmark metrics + IMPACT compatibility report
03 / Manufacturing context	MoSPI ASUSE 2023–24 + Ministry of MSME RAMP
04 / Alternatives	Cognex, Tulip and Siemens Inspekto official materials
05 / Visual assets	Founder-supplied REQUIEM; Andrea Piacquadio / Pexels
06 / Investor structure	Sequoia business-plan guidance; official Speedrun FAQ

Founder biography is attributed to the founder; no independent award certificates or factory deployment evidence were supplied.